

Algorithms Problem Solving

Level 2

Copy Odd Numbers To New Array

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Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It checks each element to determine whether it is odd or even.
- ◆ Only the odd numbers are copied into a second array.
- ◆ The copying is done using `AddElementToArray()`.
- ◆ Finally, the original array and the new array containing only odd numbers are printed.

Explanation

- ◆ `CheckOddOrEven()` determines whether a number is odd or even.
- ◆ It uses the remainder operator `%`:
 `Number % 2 != 0`
- ◆ If the remainder is not `0`, the number is odd.
- ◆ `CopyOddNumbers()` loops through all elements of `arr1`.
- ◆ For every odd number, it calls:
 `AddElementToArray(arr1[i], arr2, arr2Length);`
- ◆ `AddElementToArray()` increases `arr2Length` and stores the number at the next available position.
- ◆ Therefore, `arr2` contains only the odd numbers from `arr1`.

Steps in main()

- 1 Create two arrays with a maximum size of `100`.
- 2 Set `arr1Length` and `arr2Length` to `0`.
- 3 Fill `arr1` with random numbers from `1` to `100`.
- 4 Print `arr1`.
- 5 Loop through all elements of `arr1`.
- 6 Check whether each number is odd or even.
- 7 If the number is odd, call `AddElementToArray()`.
- 8 Add the odd number to `arr2`.
- 9 Increase `arr2Length`.
- 10 Print `arr2`.

💡 Example

◆ `Array 1` :

42 68 35 1 70 25 79 59 63 65

◆ The program checks each number:

42 → Even

68 → Even

35 → Odd

1 → Odd

70 → Even

25 → Odd

79 → Odd

59 → Odd

63 → Odd

65 → Odd

◆ Only the odd numbers are added to `arr2` :

35 1 25 79 59 63 65

➡ Result

Array 1 Elements: 42 68 35 1 70 25 79 59 63 65

Array 2 Odd Numbers: 35 1 25 79 59 63 65

Note

◆ The enum:

```
enum enOddOrEven { Odd = 1, Even = 2 };
```

◆ makes the result of `CheckOddOrEven()` easier to understand.

◆ `CheckOddOrEven()` returns `Odd` when the number is odd and `Even` when the number is even.

◆ `CopyOddNumbers()` is responsible for filtering the array.

◆ `AddElementToArray()` is responsible for adding the selected number to the second array.

◆ This separates the responsibilities of the functions.

◆ `arr2Length` starts at `0` because `arr2` initially has no elements.

⊘ Warning

- ◆ The program should ensure that the entered number of elements does not exceed `100`.
- ◆ `arr2` also has a maximum capacity of `100`.
- ◆ `ReadArrayElements()` is not used in this program.
- ◆ `cmath` and `corecrt\_math\_defines.h` are unnecessary.
- ◆ `srand()` and `time()` should ideally have their proper headers included explicitly.
- ◆ `CheckOddOrEven()` can be simplified to a `bool` function if the program only needs `true` or `false`, but the enum makes the odd/even result explicit.



Comparison

`arr1`

- ✓ Contains all the original random numbers.
- ✓ Contains both odd and even numbers.
- ✓ Acts as the source array.

`arr2`

- ✓ Starts empty.
- ✓ Contains only odd numbers from `arr1`.
- ✓ Uses `AddElementToArray()` to add each selected number.
- ✓ Its length can be smaller than `arr1Length`.

✦ Summary

- ✓ Fill the first array with random numbers.
- ✓ Check every element using `CheckOddOrEven()`.
- ✓ Select only odd numbers.
- ✓ Use `AddElementToArray()` to add them to the second array.
- ✓ `arr2Length` keeps track of how many odd numbers were copied.
- ✓ The second array contains only the odd numbers from the first array.

Thank you 😊